COMP4442 Service and Cloud Computing
Lab 1

Weiyang MA
weiyang.ma@connect.polyu.hk

Feb. 04 2026

Outline

- Create Your AWS Education Account (15 mins)
- How to use AWS (35 mins)
  - Access the AWS console
  - Search for the AWS services
  - A simple example
- Q&A

Create Your AWS Education Account

- AWS Education enables students to register account without credit card
- This year, students cannot register the education account by themselves (invitation from educators is required)
- Check your email

Create Your AWS Education Account

- Click the link on your email, and go to the sign in page

AA

AWS Academy<notifications@instructure.com>

To: MA, Weiyang [Student]

Tue 2026-0

CAUTION: This email is not originated from PolyU. Do not click links or open attachments unless recognize the sender and know the content is safe.

You've been invited to participate in a class at AWS Academy . The class is called AWS Academy Learner Lab [153063]. Course role: TA

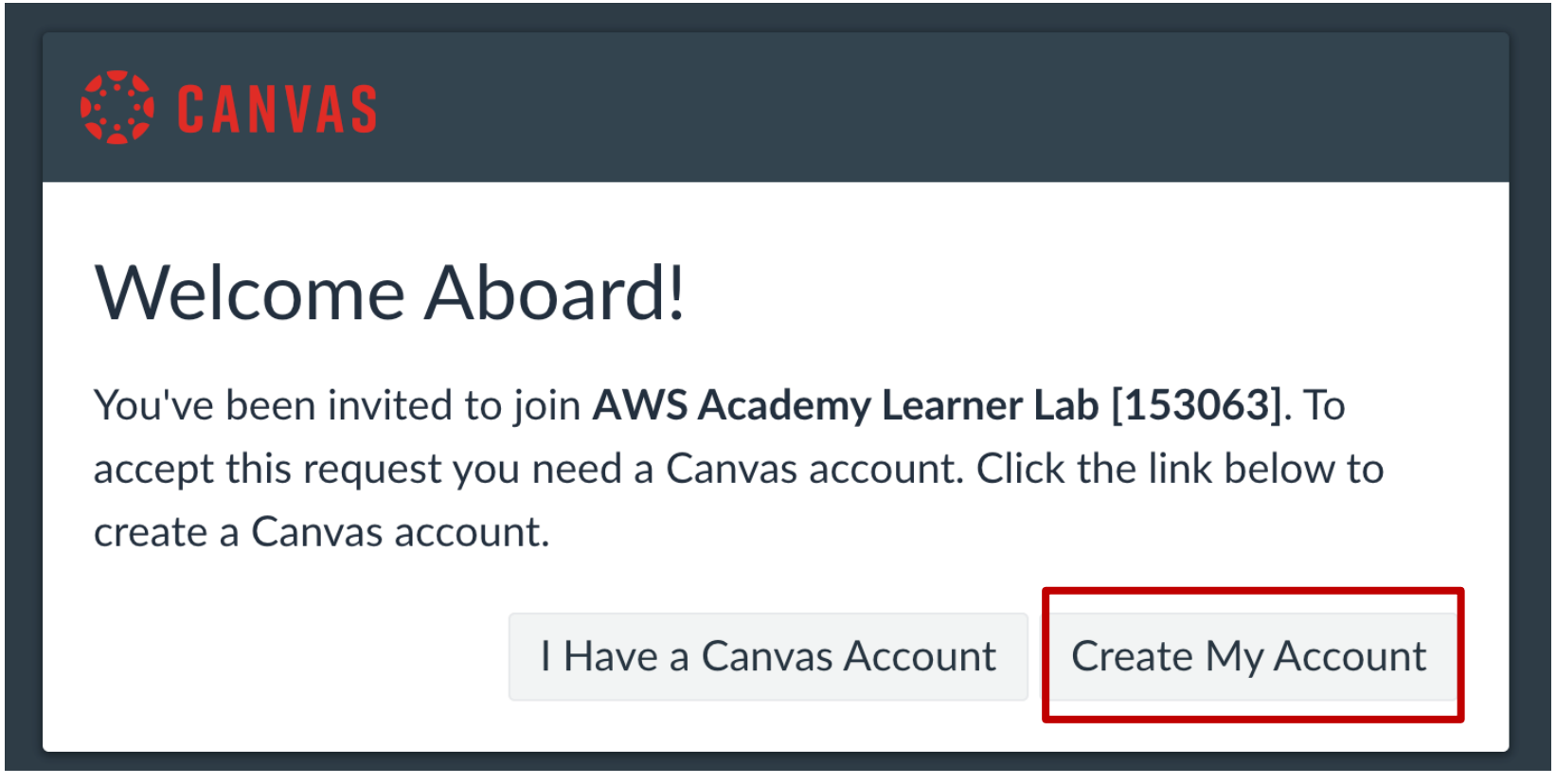
Name: **25034047r@connect.polyu.hk**
Email: **25034047r@connect.polyu.hk**
Username: **none**

You'll need to register with Canvas before you can participate in the class.

Get Started

Create Your AWS Education Account

- Click the link on your email, and go to the sign in page



Create Your AWS Education Account

- Now set your password and login in
- https://awsacademy.instructure.com/

Welcome Aboard!

In order to finish signing you up for the course **AWS Academy Learner Lab [153063]**, we'll need a little more information.

Login: 25034047r@connect.polyu

Password: * ••••••••••••

Time Zone: Hong Kong (+08:00)

aws academy

Student Login

(For students enrolled in a class)

学生の方はこちらからログインしてください。

已注册课程的学生请在这里登录

Educator Login

(For educators who have access to the AWS Academy Portal)

講師の方（AWS Academyメンバーポータルのアカウントをお持ちの方）はこちらからログインしてください。

教师请在这里登录（您需使用AWS Academy Portal账户登录）

Create Your AWS Education Account

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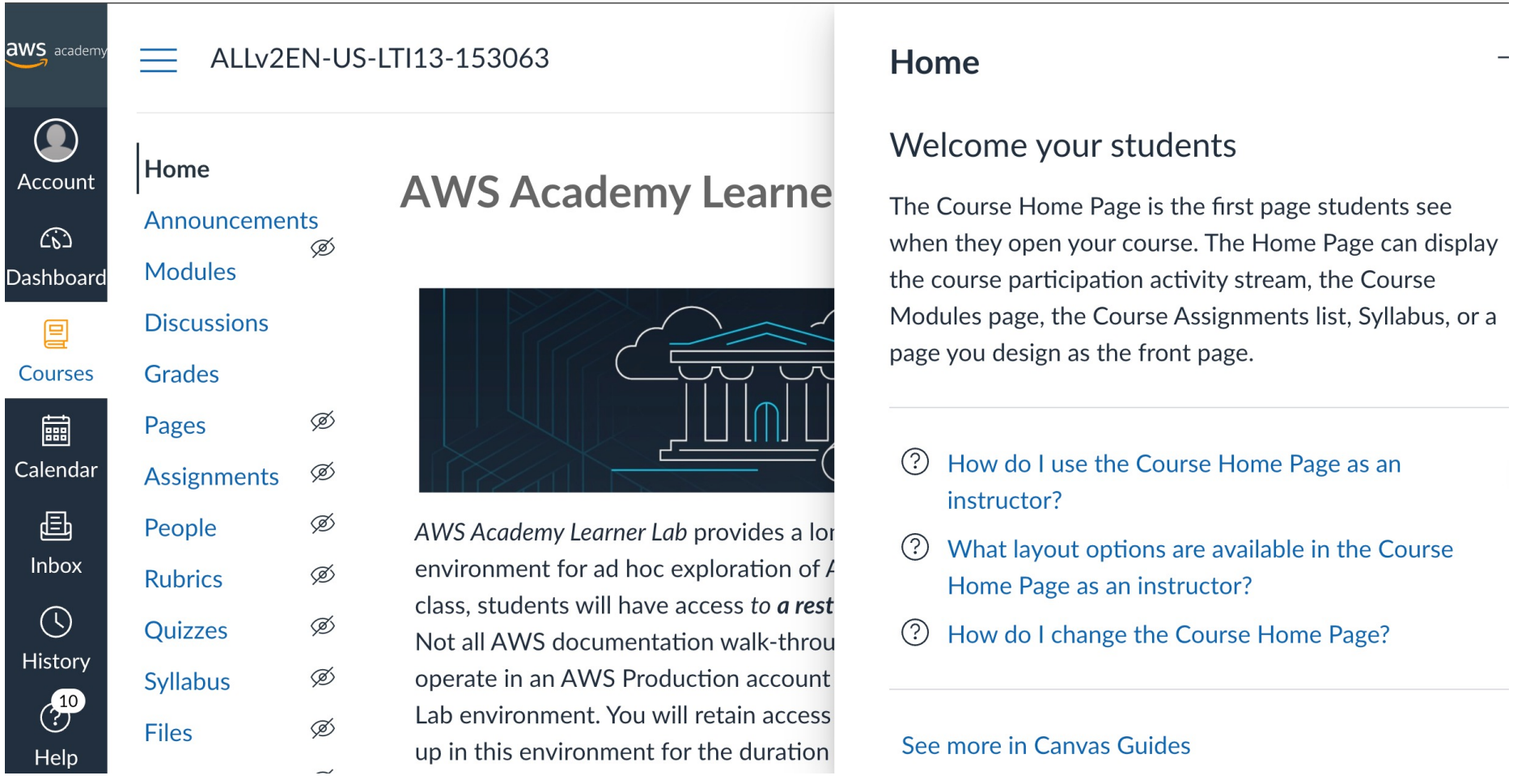
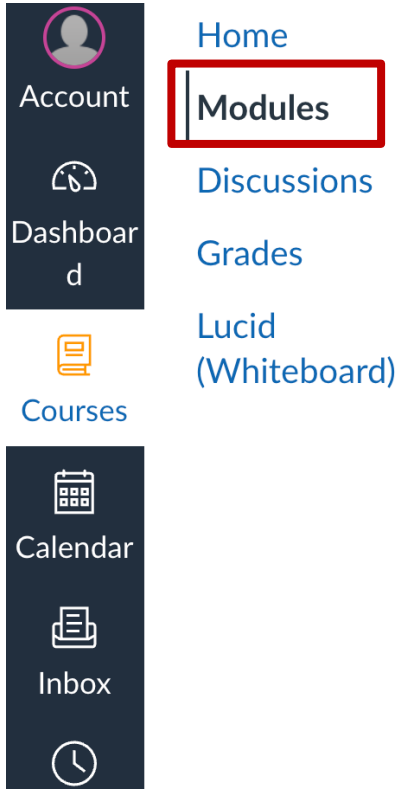
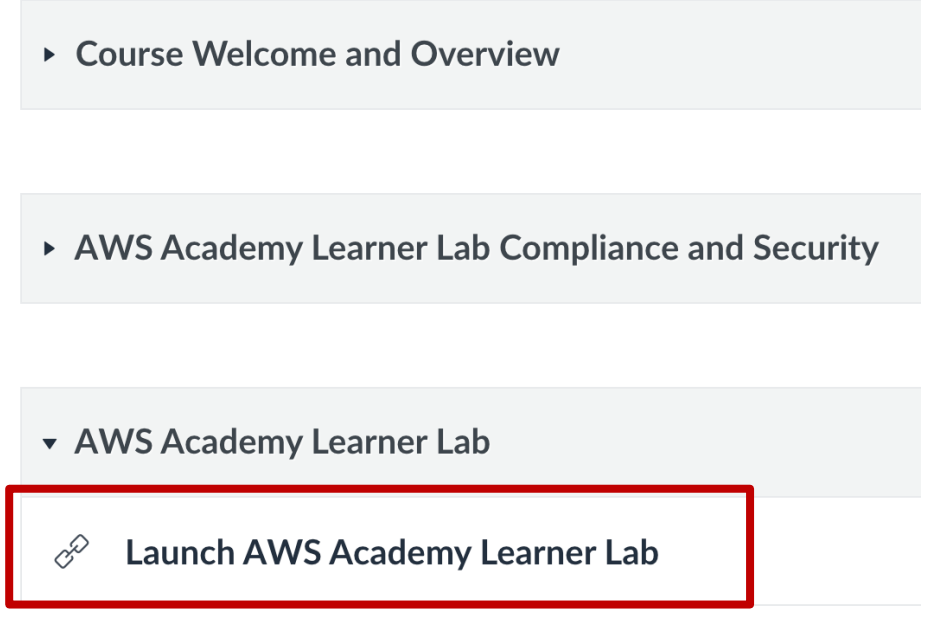
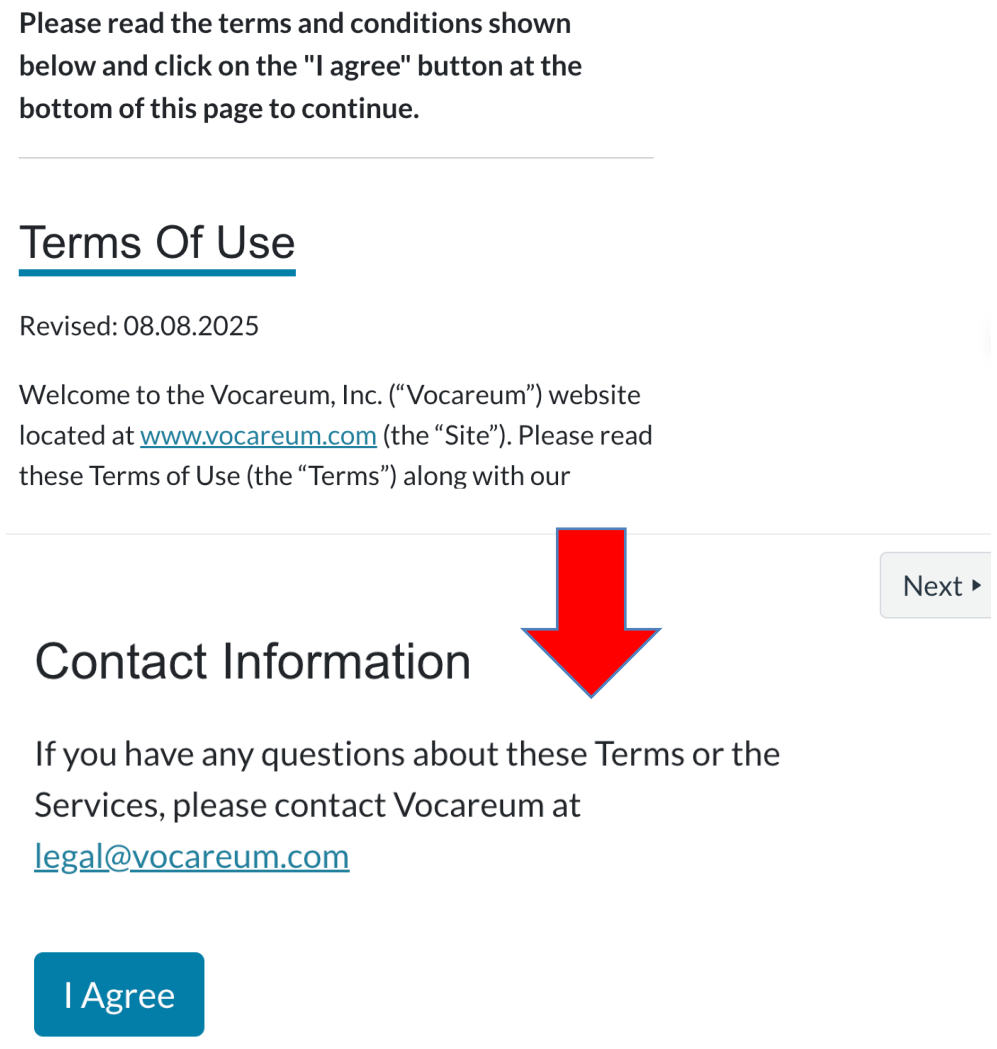


How to Use AWS

- Access the AWS console

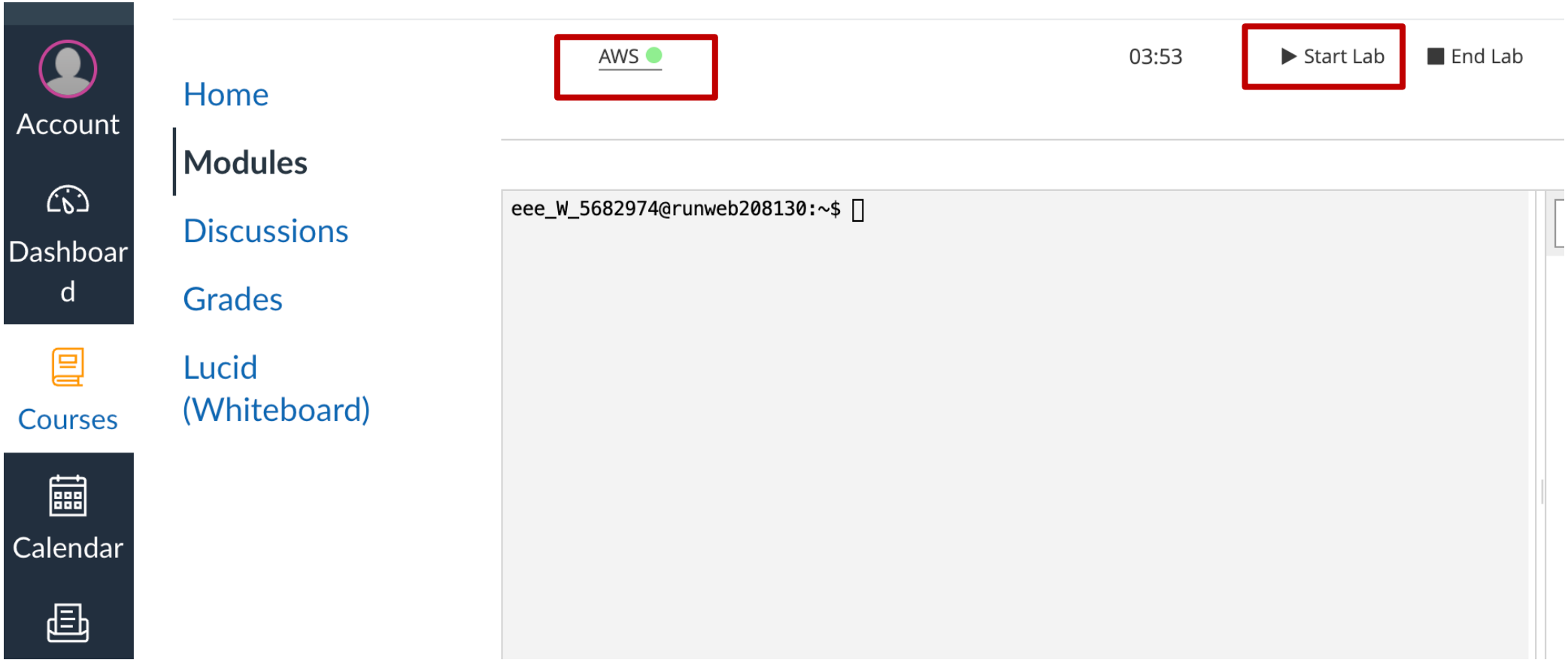






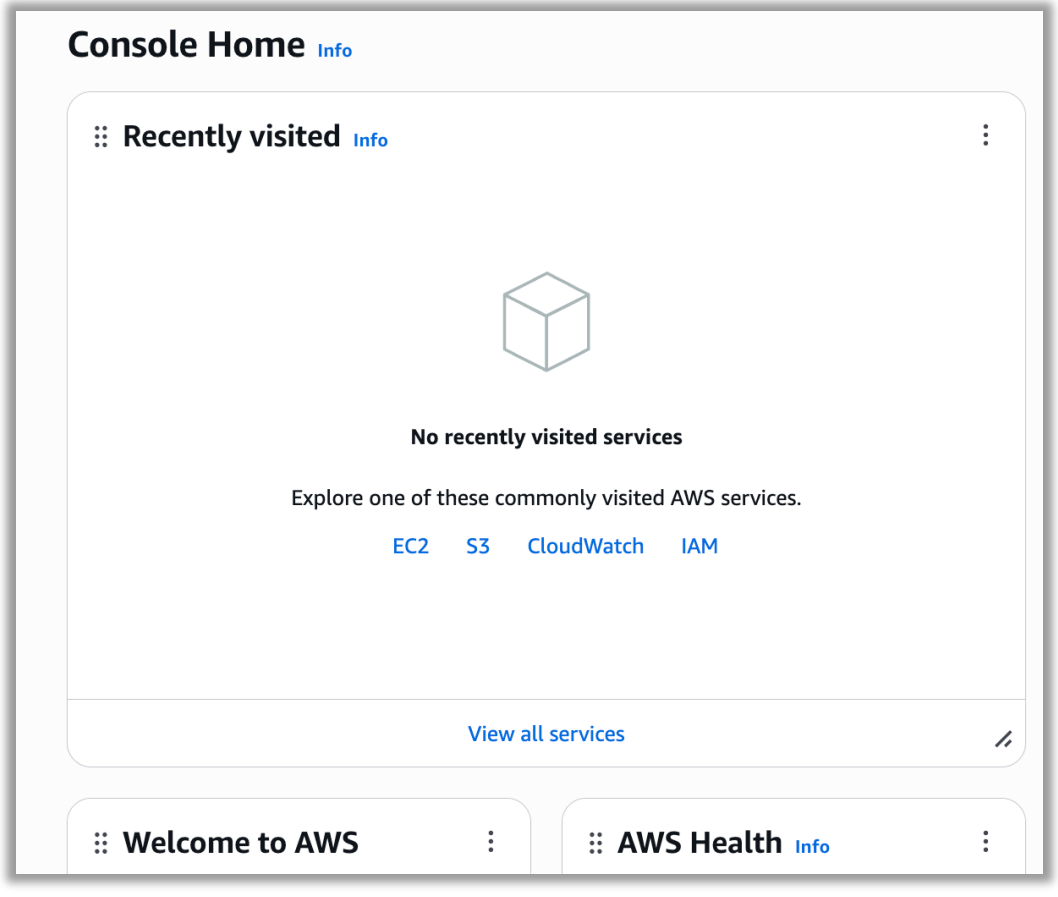
How to Use AWS

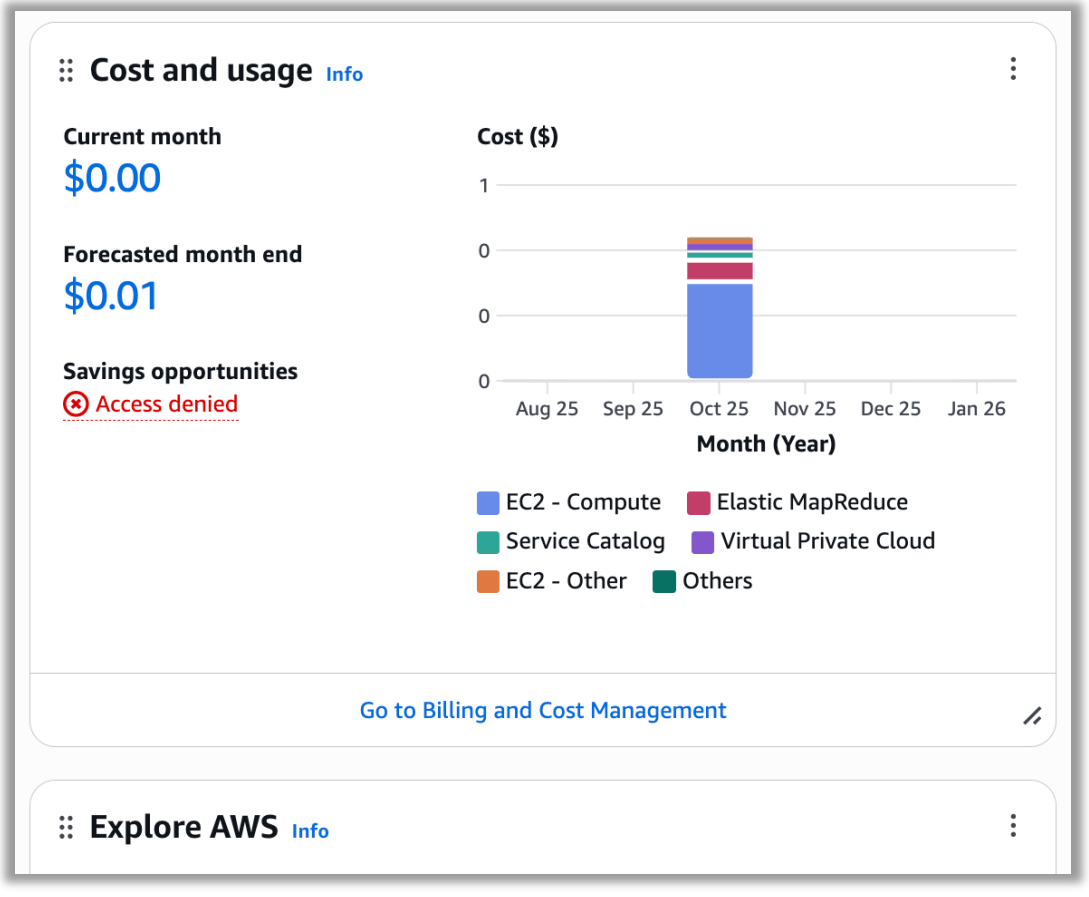
- Access the AWS console



How to Use AWS

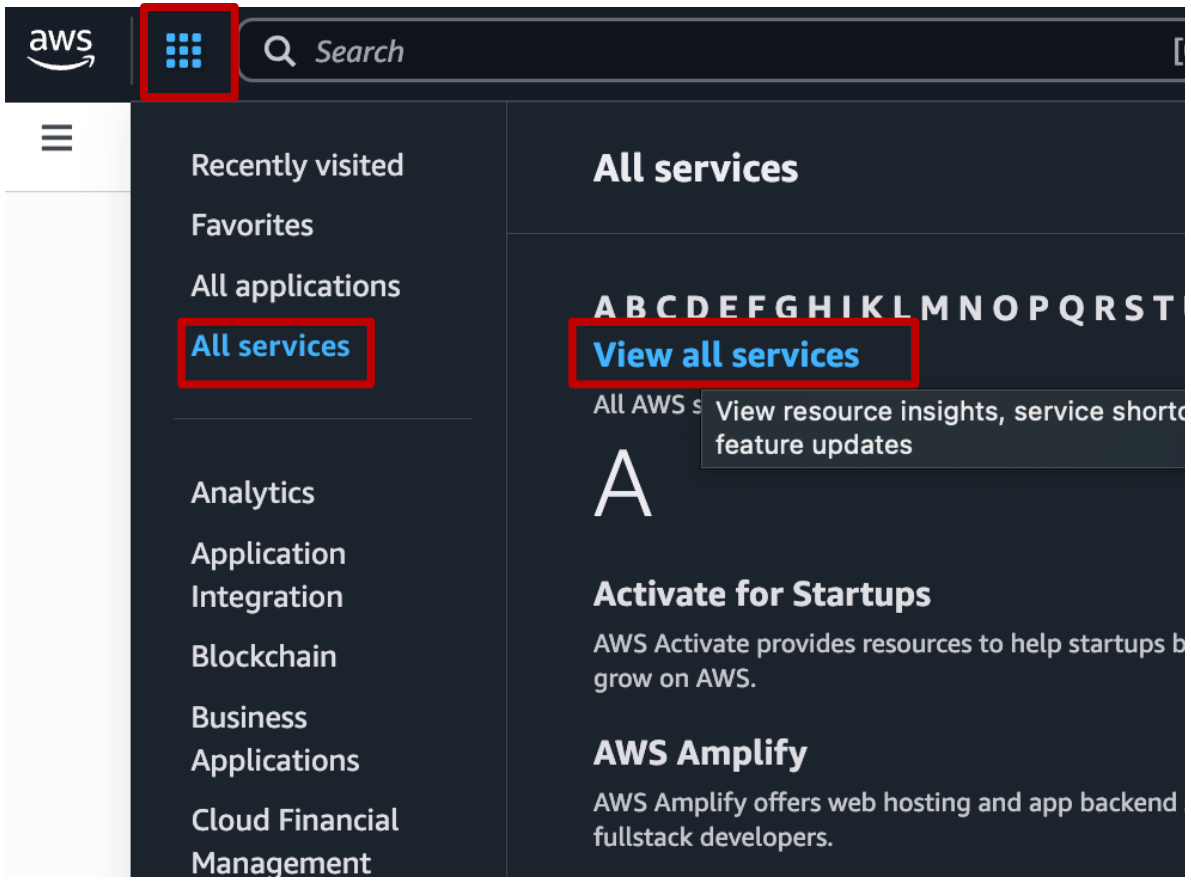
- AWS console

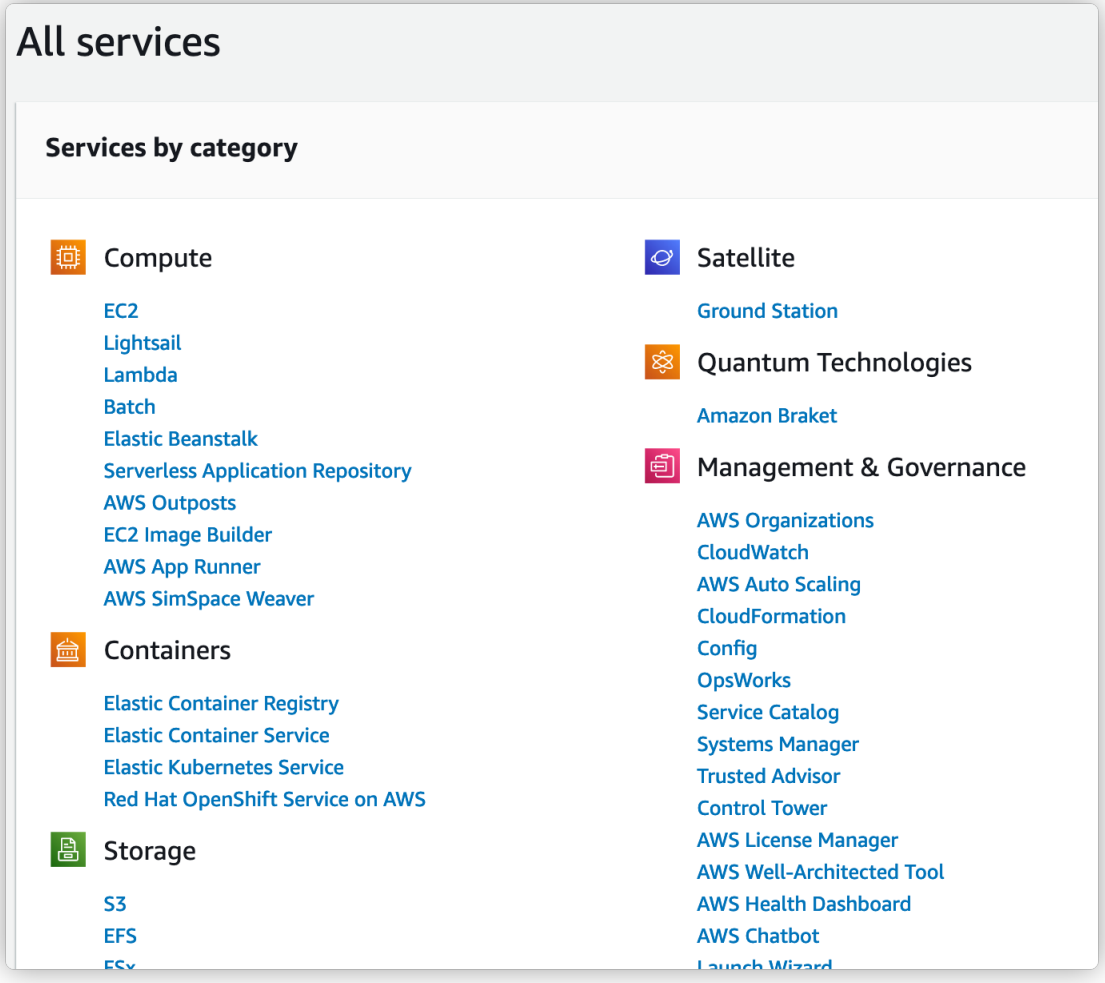




How to Use AWS

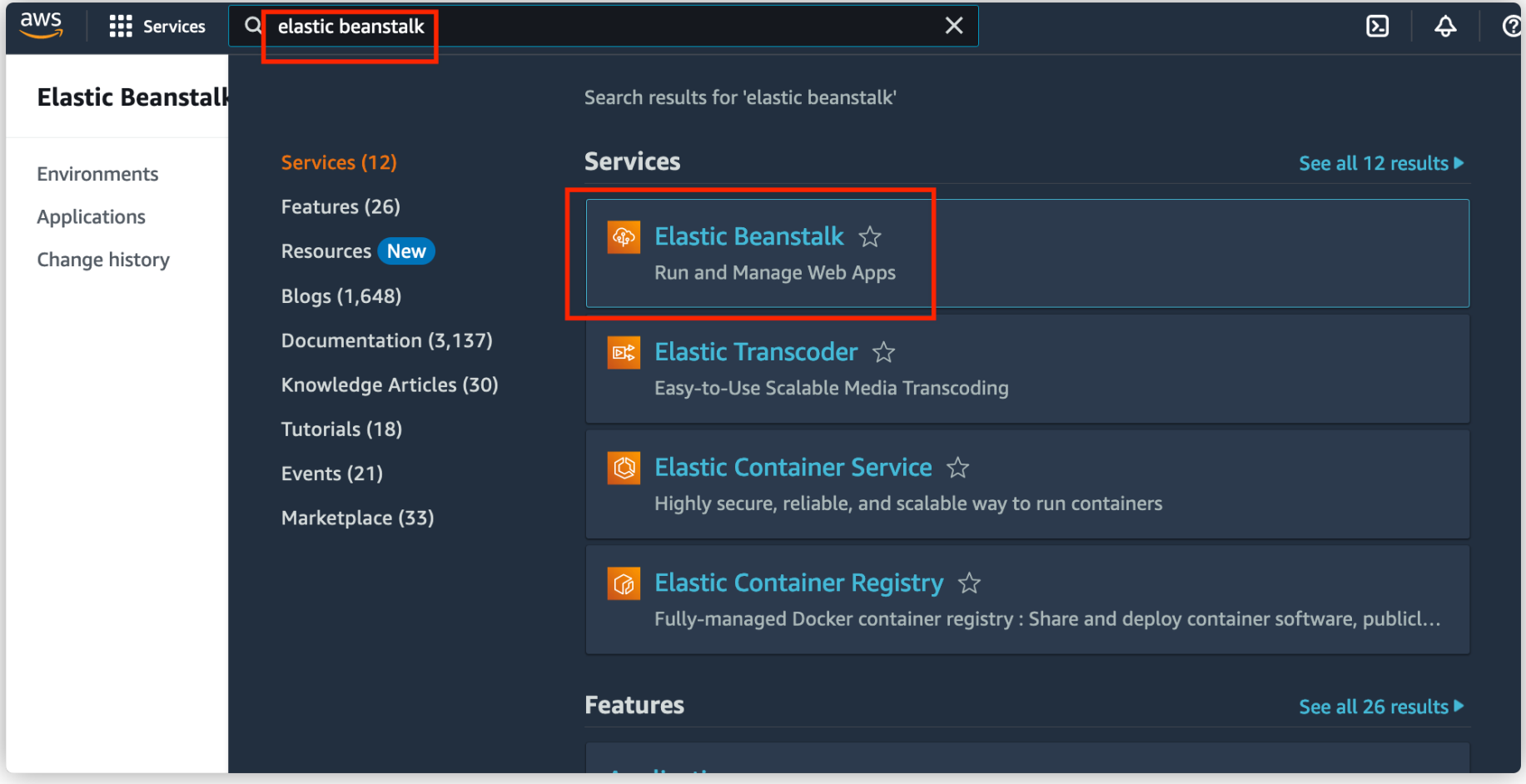
- Access AWS services





How to Use AWS

- Access AWS services



A Simple Example

- Create a web app

Compute

Amazon Elastic Beanstalk

End-to-end web application management.

Amazon Elastic Beanstalk is an easy-to-use service for deploying and scaling web applications and services developed with Java, .NET, PHP, Node.js, Python, Ruby, Go, and Docker on familiar servers such as Apache, Nginx, Passenger, and IIS.

Get started

Easily deploy your web application in minutes.

Create Application

Pricing

There's no additional charge for Elastic Beanstalk. You pay for Amazon Web

How it works

Configure environment Info

Environment tier Info

Amazon Elastic Beanstalk has two types of environment tiers to support different types of web applications.

- (●) Web server environment
  Run a website, web application, or web API that serves HTTP requests. Learn more
- ( ) Worker environment
  Run a worker application that processes long-running workloads on demand or performs tasks on a schedule. Learn more

Application information Info

Application name

comp4442-lab1

Maximum length of 100 characters.

Platform Info

Platform	Platform branch	Platform version
Python	Python 3.9 running on 64bit Amazon Linux 2023	4.9.1 (Recommended)

A Simple Example

- Create a web app

Application code Info

- (●) Sample application
- ( ) Existing version
  Application versions that you have uploaded.
- ( ) Upload your code
  Upload a source bundle from your computer or copy one from Amazon S3.

Presets Info

Start from a preset that matches your use case or choose custom configuration to unset

Configuration presets

- (●) Single instance (free tier eligible)
- ( ) Single instance (using spot instance)
- ( ) High availability
- ( ) High availability (using spot and on-demand instances)
- ( ) Custom configuration

Cancel Next

A Simple Example

- Create a web app

Configure service access Info

Service access

IAM roles, assumed by Elastic Beanstalk as a service role, and EC2 instance profiles allow Elastic Beanstalk to create and manage your environment. Both the and instance profile must be attached to IAM managed policies that contain the required permissions. Learn more

Service role

Choose an IAM role for Elastic Beanstalk to assume as a service role. The IAM role must have the required IAM managed policies.

LabRole | Create role

EC2 instance profile

Choose an IAM instance profile with managed policies that allow your EC2 instances to perform required operations.

LabInstanceProfile | Create role

EC2 key pair - *optional*

Select an EC2 key pair to securely log in to your EC2 instances. Learn more

vockey

Cancel | Skip to review | Previous

A Simple Example

- Create a web app

Review Info

Step 1: Configure environment

Environment information

Environment tier
Web server environment

Application name
test

Environment name
Test-env

Application code
Sample application

Platform
arn:aws:elasticbeanstalk:us-east-1::platform/Python 3.9 running on 64bit Amazon Linux 2023/4.9.2

NumThreads
15

WSGIPath
application

Logs retention
7

Rotate logs
Disabled

minor

X-Ray enabled
Disabled

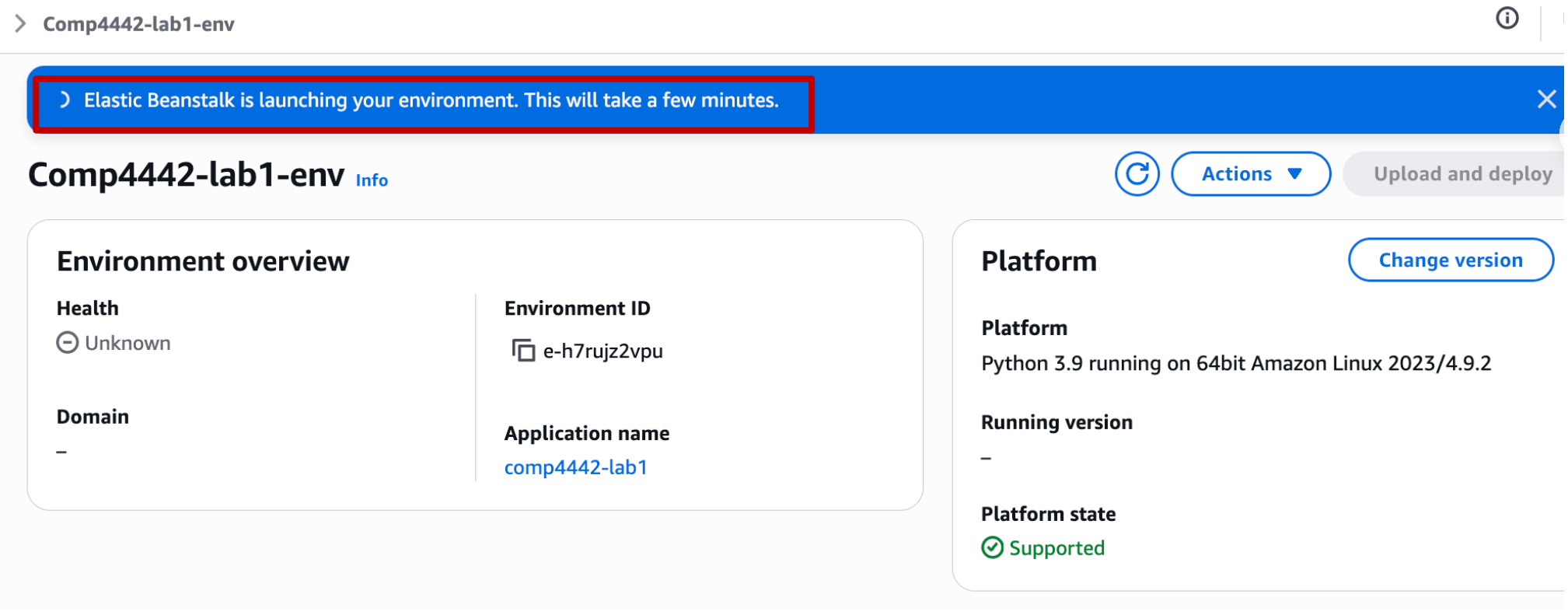
Environment properties

Source	Key	Value
Plain text	PYTHONPATH	/var/app/venv/staging-LQM1lest/bin

Cancel Previous Create

A Simple Example

- Wait for a moment…



A Simple Example

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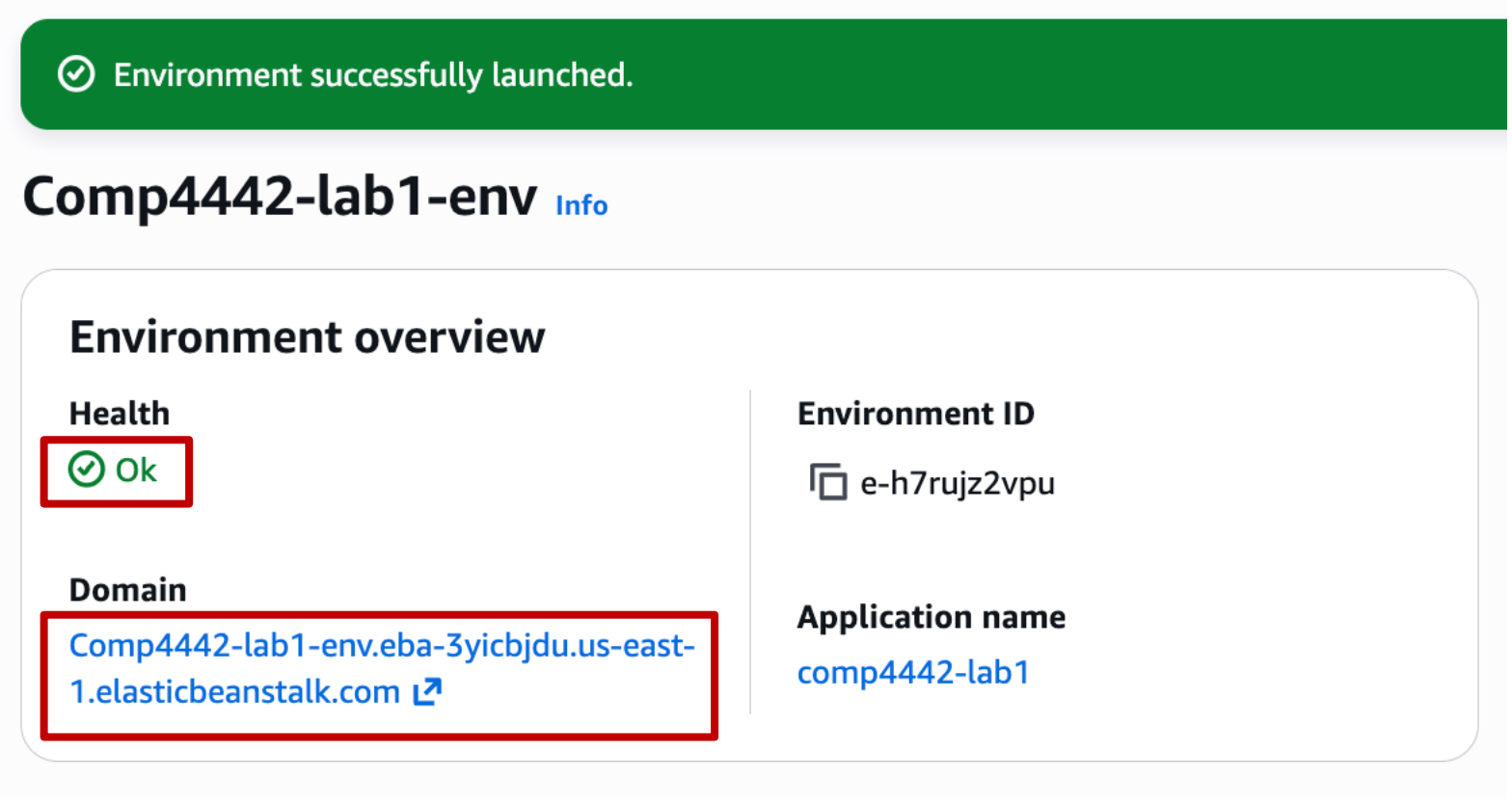
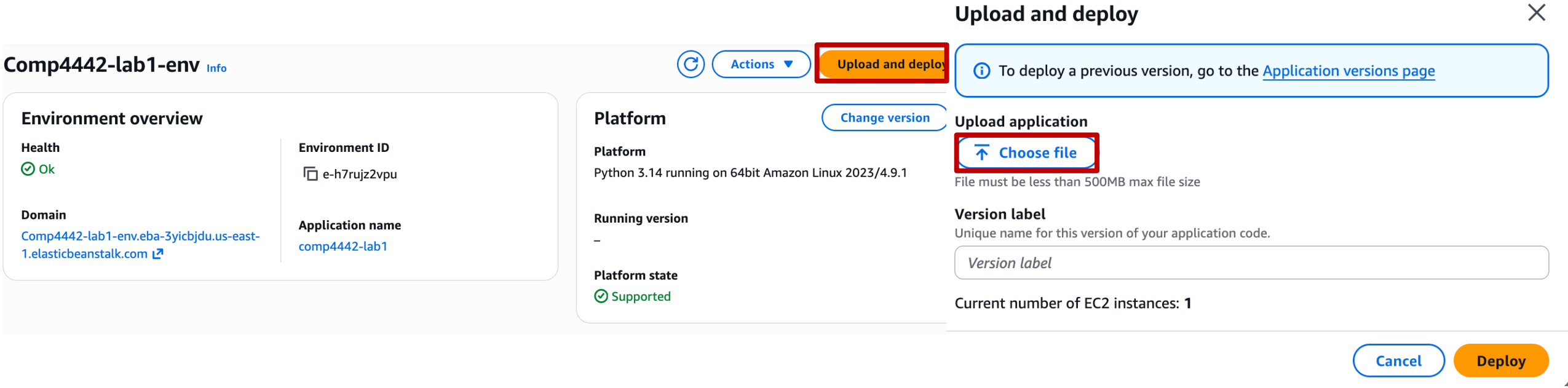


A Simple Example

- Access the default website

A Simple Example

- Upload the code (find the `flask_app.zip` file in the Blackboard)



A Simple Example

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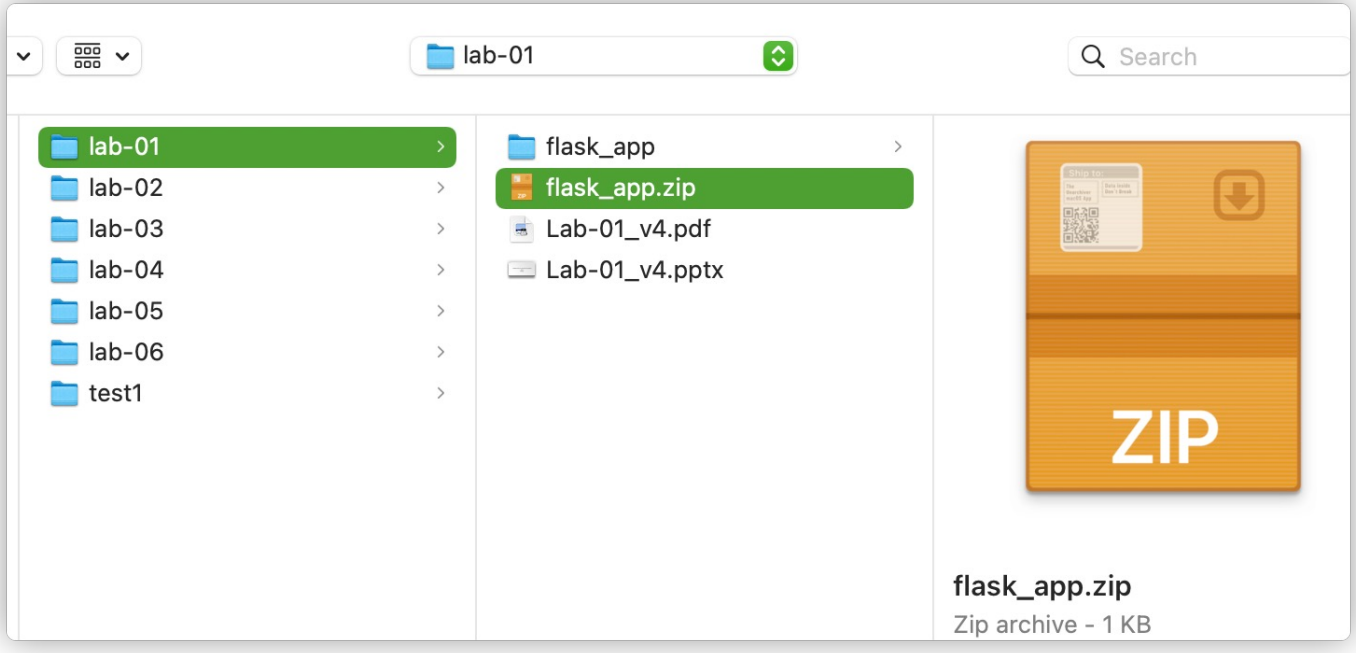
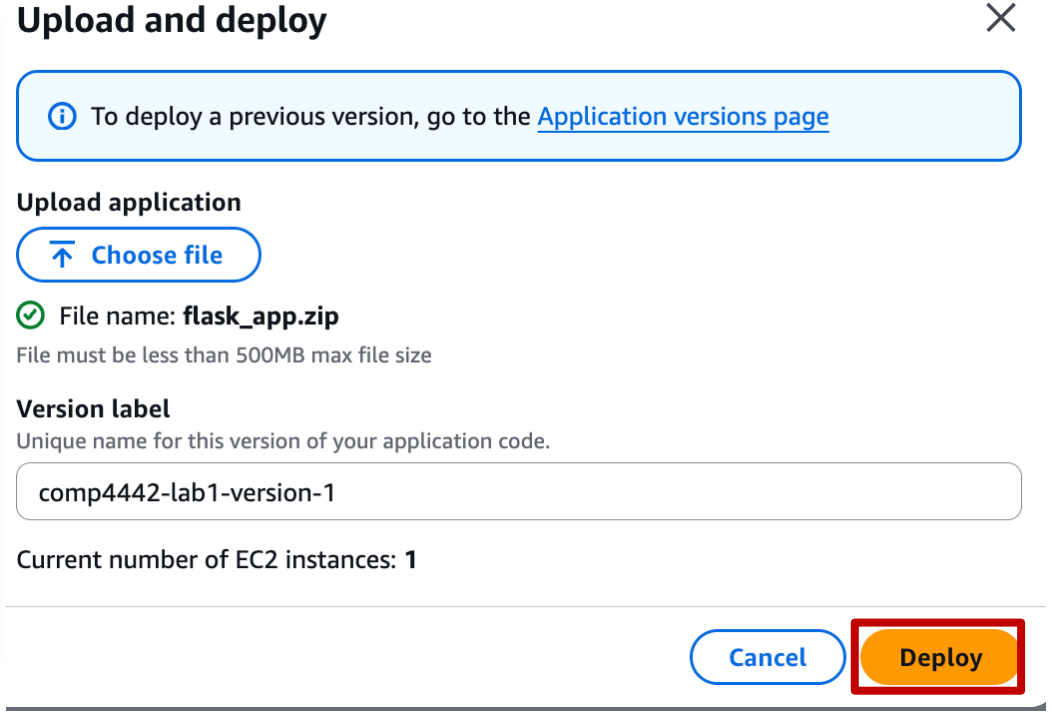
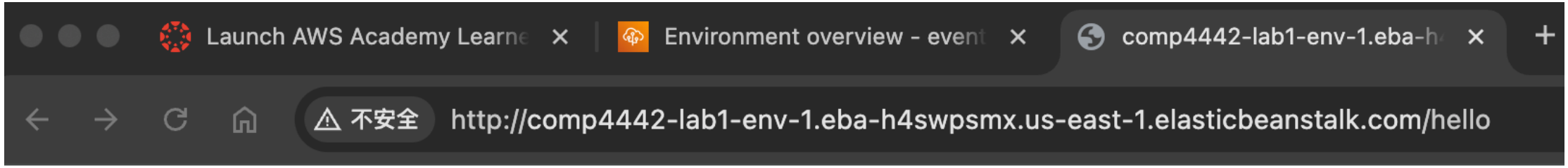




A Simple Example

- Hello world!
  - Access the website via `<your domain>/hello`

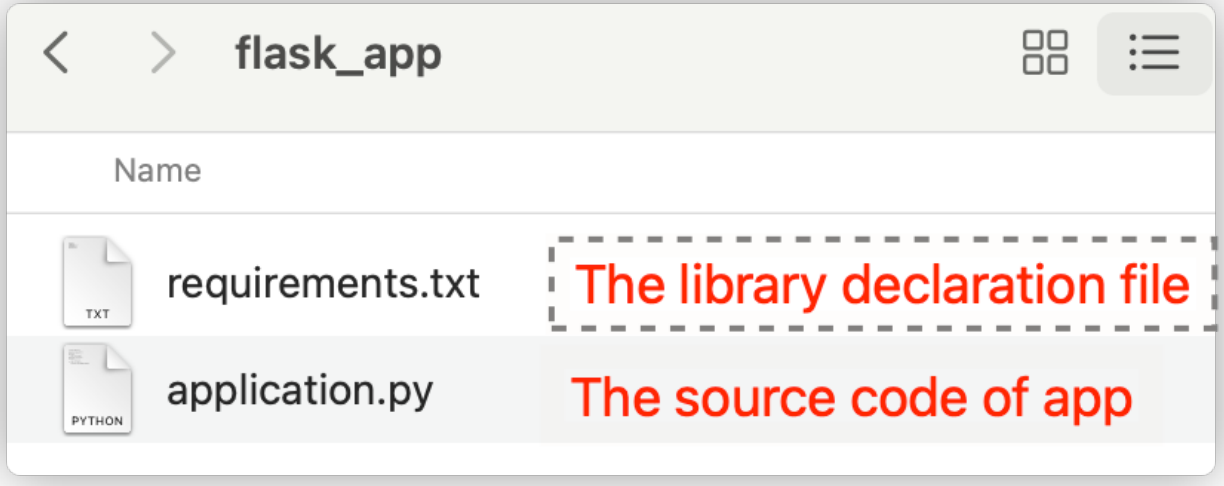


Hello World!

A Simple Example

- The content of flask_app.zip

$pip freeze > requirements.txt




```python
from flask import Flask

application = Flask(__name__)

@application.route("/")
def index():
    return "Your Flask App Works!"

@application.route("/hello")
def hello():
    return "Hello World!"


if __name__ == "__main__":
    application.run(port=5000, debug=True)
```

Q&A